Concise Building Vibration Assessment

A peak-centered comparison for spectroscopy and other vibration-sensitive work

Source workbook: `FinalReportData.xlsx`
Measurements: 15 spectra from 5 measured rooms
Report date: July 24, 2026

Simplified companion to the full building-vibration report

1 Summary

Recommendation: use the Titan laboratory. Titan has the lowest estimated broadband vibration background among the candidate rooms and the cleanest spectrum away from one narrow line at approximately 29.8 Hz. PHYS 245 is the next-best room, but it has more distinct peaks, especially near 17.6 Hz, 29.8 Hz, and 66.0 Hz.

For peak-sensitive spectroscopy, the important distinction is not a small difference in the common noise floor; it is the number and size of discrete lines. Titan's median non-mains ASD is $2.24 \times 10^{-4}\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$. Its median-filtered broadband RMS estimate is $3.58 \times 10^{-3}\,\mathrm{m\,s^{-2}}$. PHYS 245 is close at $3.74 \times 10^{-3}\,\mathrm{m\,s^{-2}}$, but has three prominent non-mains peaks rather than Titan's one.

PHYS 111 and Herzog are less attractive. Both retain a strong 29.8 Hz line after the freezer-informed adjustment, reaching $1.80 \times 10^{-2}\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$ in PHYS 111 and $2.29 \times 10^{-2}\,\mathrm{m\,s^{-2}\,Hz^{-0.5}}$ in Herzog. These are approximately 3.5 and 4.4 times the corresponding Titan peak.

Early note on 60 Hz. A narrow feature near 60.0 Hz is present in every location and is very likely influenced by instrumental coupling to electrical mains. It is excluded from the room comparison. Its amplitude is not constant, so a mechanical or equipment-dependent contribution may also be present.

2 Analysis approach

The analysis uses the numerical frequency and acceleration-ASD columns from which the workbook plots were made. The separate peak lists printed in the worksheets are not used for peak detection.

- Peak finding is performed on the unsmoothed numerical ASD.
- The first 2.00 Hz is omitted, providing margin beyond the DC bin.
- The entire 58.0 Hz to 62.0 Hz band is omitted from peak and RMS comparisons.
- Integrated RMS is calculated from 2.00 Hz to 250 Hz. Because narrow lines can dominate this value, a median-filtered broadband estimate and the individual peaks are shown separately.
- The spectra are single measurements rather than repeated trials, so small differences should not be overinterpreted.

3 What the floor spectra show

The broad noise floors are similar, particularly at higher frequencies. Room selection is therefore driven mainly by the discrete peaks. The integrated values in Figure 2 demonstrate why a simple RMS ranking can be misleading: Titan's 29.8 Hz line raises its total, even though the background between peaks is the lowest of the candidate rooms.

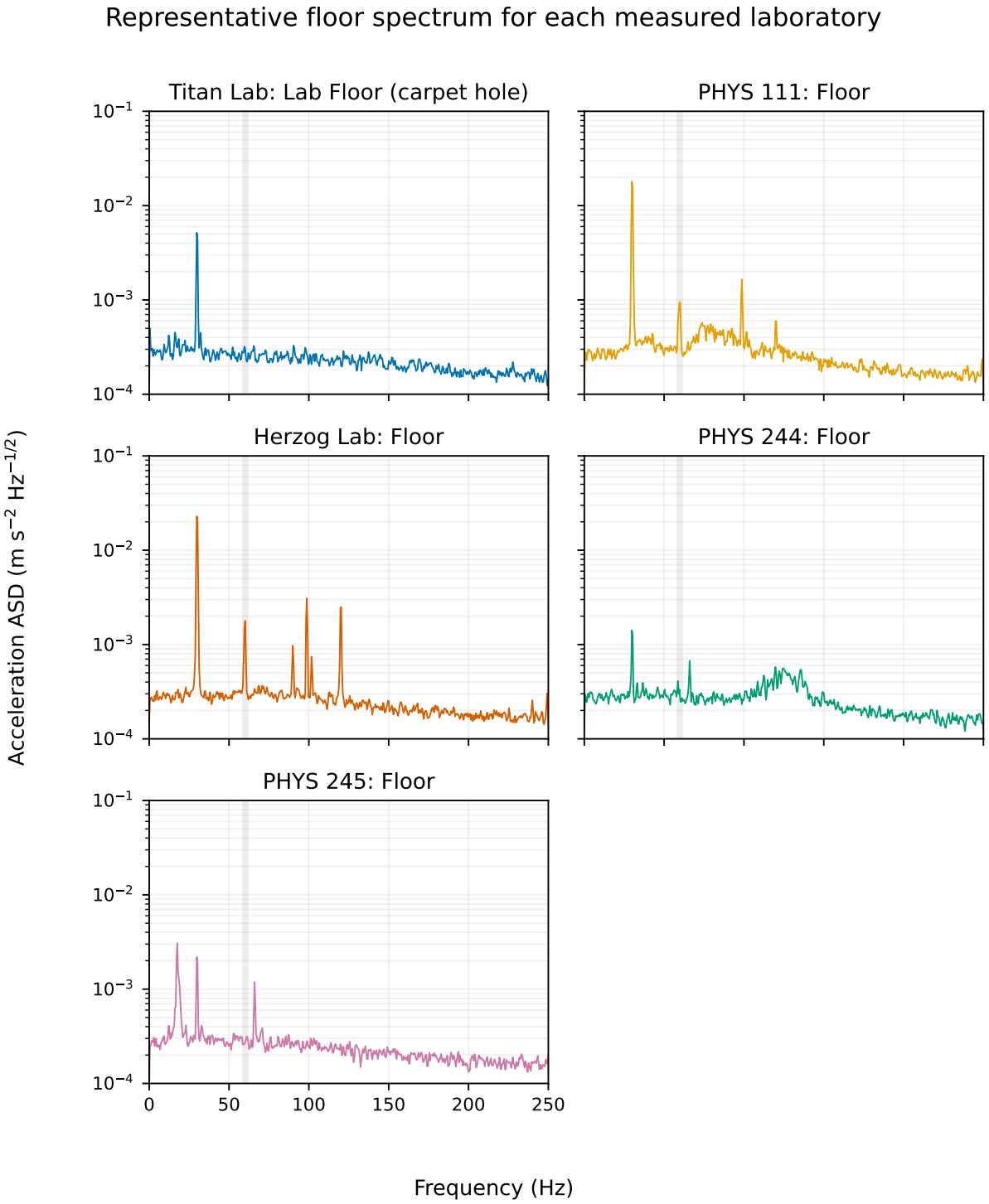


Figure 1: Representative floor spectra, shown separately with common axes. The gray band is the excluded 58.0 Hz to 62.0 Hz mains region. This is the same floor-by-floor presentation used in the full report.

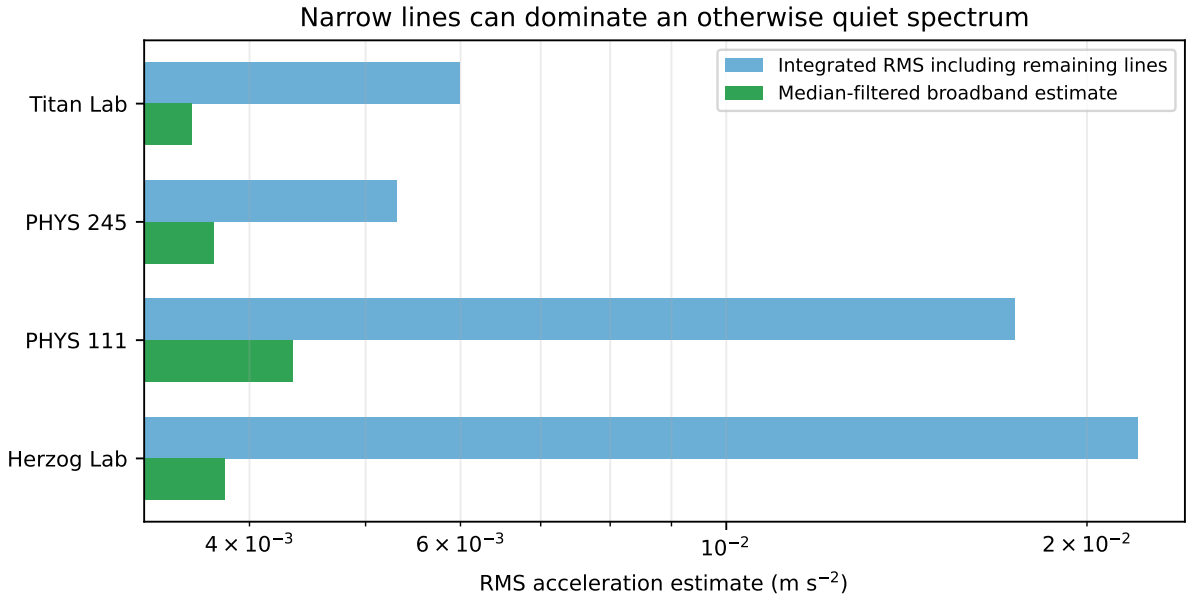


Figure 2: Integrated RMS including the remaining narrow lines compared with a median-filtered estimate of the broadband background. The PHYS 111 and Herzog integrated values use the conservative freezer adjustment described below.

Table 1: Peak-centered comparison of the candidate floor measurements. RMS values are in $10^{-3}\ \mathrm{m\,s^{-2}}$; peak ASD values are in $10^{-3}\ \mathrm{m\,s^{-2}\,Hz^{-0.5}}$.

Room	Integrated RMS	Background estimate	Largest peak	Peak ASD
Titan Lab	5.99	3.58	29.8 Hz	5.15
PHYS 245	5.31	3.74	17.6 Hz	3.07
PHYS 111	17.4	4.35	29.8 Hz	18.0
Herzog Lab	22.0	3.81	29.8 Hz	22.9

3.1 PHYS 245 and its neighboring room

The PHYS 245 and PHYS 244 floor measurements are similar in broadband level, which is reasonable given their proximity. PHYS 245 actually has the slightly lower median-filtered background estimate ($3.74 \times 10^{-3}\,\mathrm{m\,s^{-2}}$ versus $4.18 \times 10^{-3}\,\mathrm{m\,s^{-2}}$), but the rooms are not spectrally identical. PHYS 245 has stronger narrow lines near 17.6, 29.8, and 66.0 Hz.

4 Optical-table performance

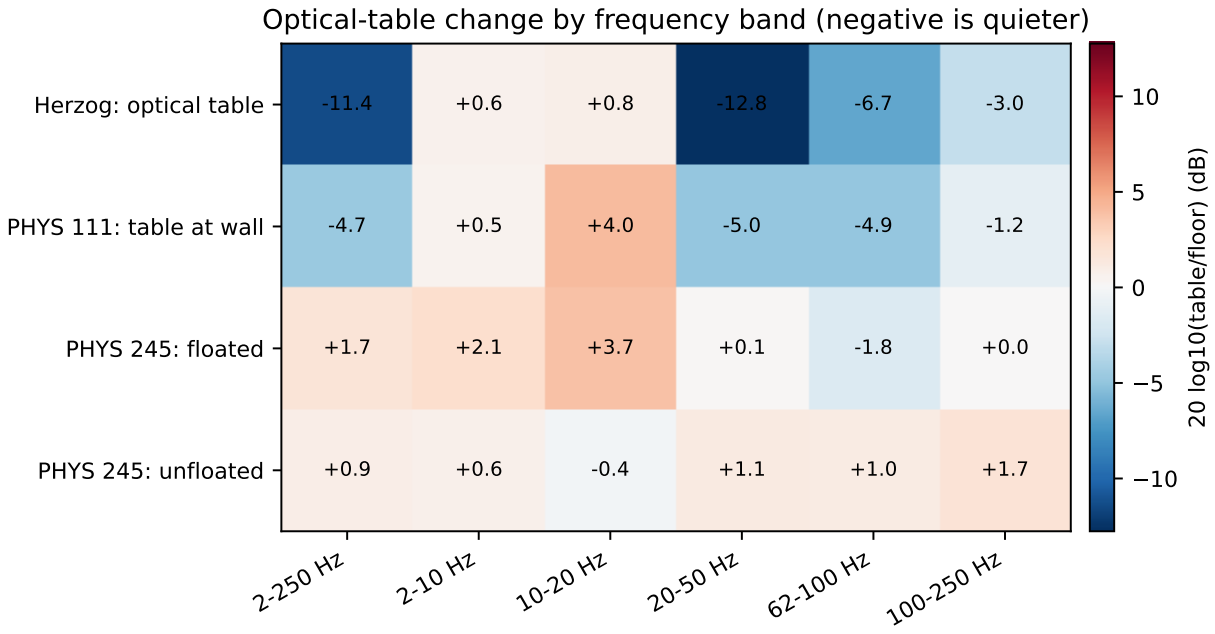


Figure 3: Band-limited change from floor to table. Negative values are attenuation; positive values are amplification.

The plot carries the main conclusion. The Herzog optical-table measurement is substantially quieter above 20.0 Hz, and the PHYS 111 table at the Herzog wall also improves the higher-frequency bands. The explicitly floated PHYS 245 table does not improve the supplied run overall and is worse from 10.0 Hz to 20.0 Hz; its isolators and loading should be checked before that result is treated as representative.

5 Herzog, PHYS 111, and the freezer

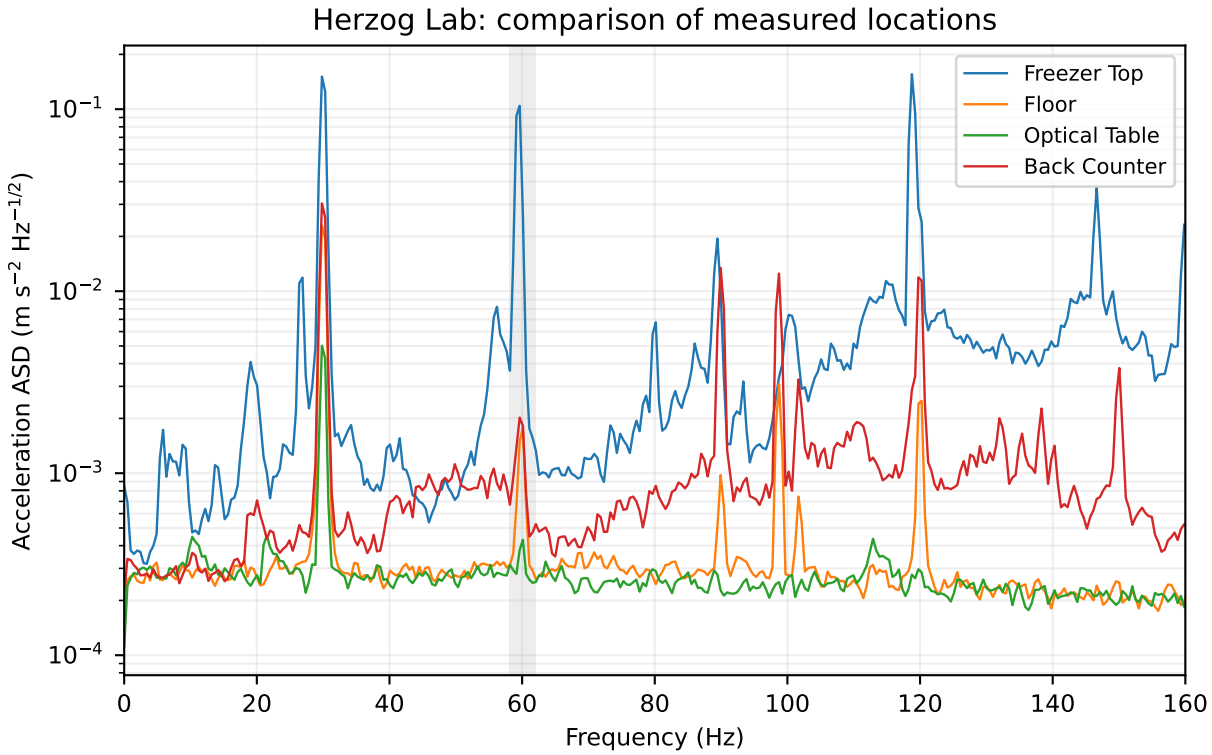


Figure 4: The measured locations in Herzog. The freezer-top run is a source reference, not a candidate experimental location.

The spectra measured on the freezer and on the Herzog floor are used as a source-reference pair. Peaks are matched using only the numerical ASD data. A matched line is treated as freezer-specific only when it is not also present on at least two distant floor spectra. This identifies a Herzog feature near 89.9 Hz that matches the freezer near 89.4 Hz, and a Herzog/PHYS 111 feature near 120 Hz that matches the freezer's strong cluster near 119 Hz.

The 29.8 Hz line is strong on the freezer, but it is also present on all distant floors. It is therefore retained rather than attributed to the freezer.

6 Why Titan is preferred

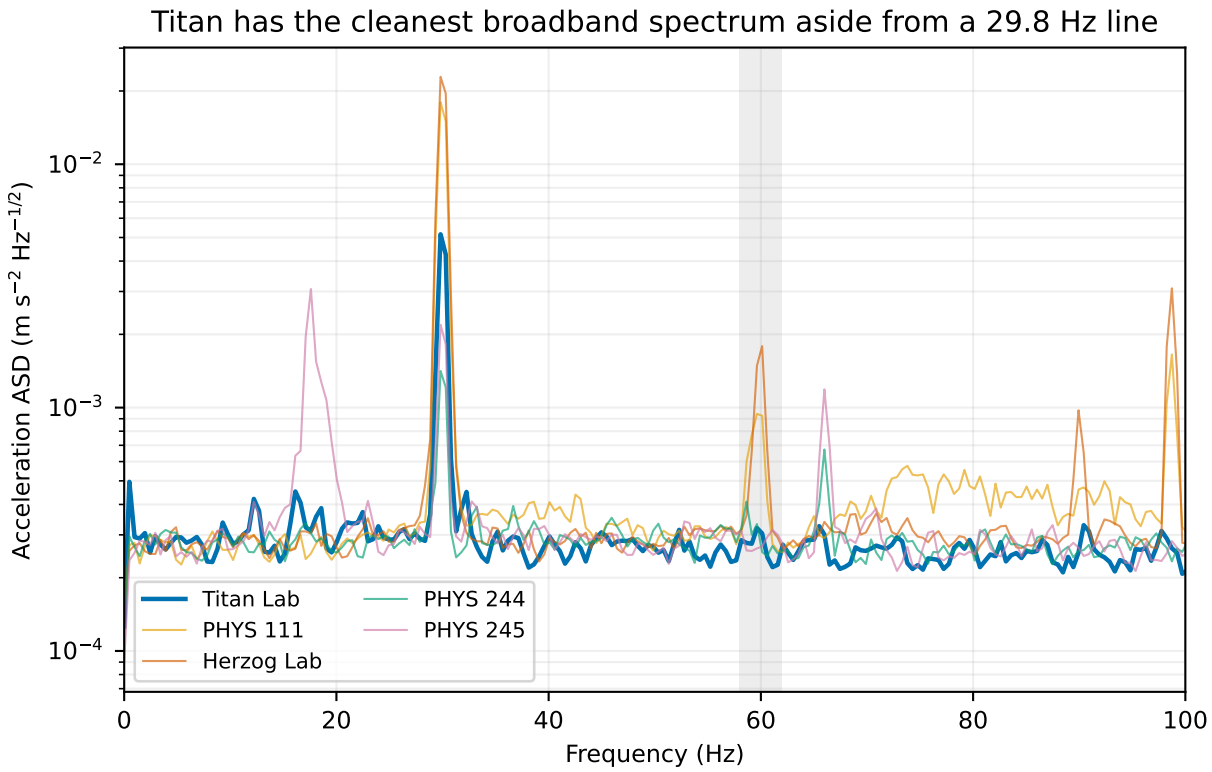


Figure 5: Titan compared with the representative floor spectra. Titan has the lowest broadband background and the fewest prominent non-mains lines. Its 29.8 Hz feature remains important for experiments with a mechanical response near that frequency.

Figure 5 is the clearest practical summary. Titan stays at or below the other candidate spectra through most of the displayed range. Its one important qualification is the narrow 29.8 Hz line. For spectroscopy, a floated and well-tuned table in Titan should therefore be the first configuration tested.

7 Freezer-informed adjustment

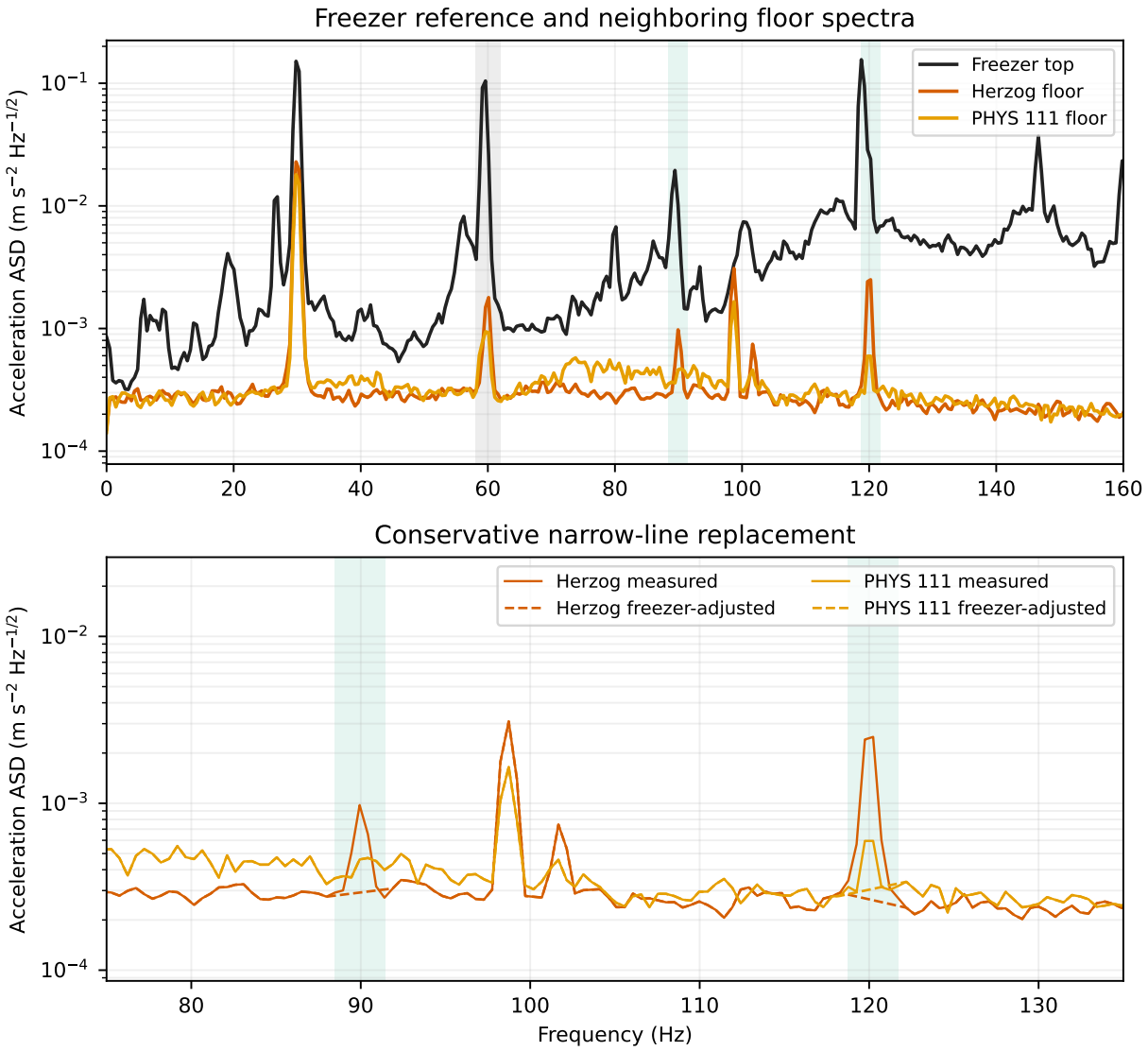


Figure 6: Top: the freezer reference with the Herzog and PHYS 111 floors. Bottom: measured and freezer-adjusted floor spectra. Green bands mark the narrow lines replaced in at least one room; the gray band marks the excluded mains region.

The adjustment does not subtract the freezer ASD directly. The measurements were not simultaneous, and no phase or transfer function is available, so direct amplitude subtraction would not be defensible. Instead, only the source-specific matched lines are replaced over a narrow $\pm 1.50\,\mathrm{Hz}$ interval by a log-linear interpolation of the local floor background.

This adjustment lowers the Herzog integrated RMS from $2.22 \times 10^{-2}\,\mathrm{m\,s^{-2}}$ to $2.20 \times 10^{-2}\,\mathrm{m\,s^{-2}}$; the PHYS 111 value changes from $1.74 \times 10^{-2}\,\mathrm{m\,s^{-2}}$ to $1.74 \times 10^{-2}\,\mathrm{m\,s^{-2}}$ at the shown precision. The small change confirms that the freezer-specific lines are real but are not the main reason these rooms compare poorly. The retained 29.8 Hz line is the dominant difference.

8 The 60 Hz feature

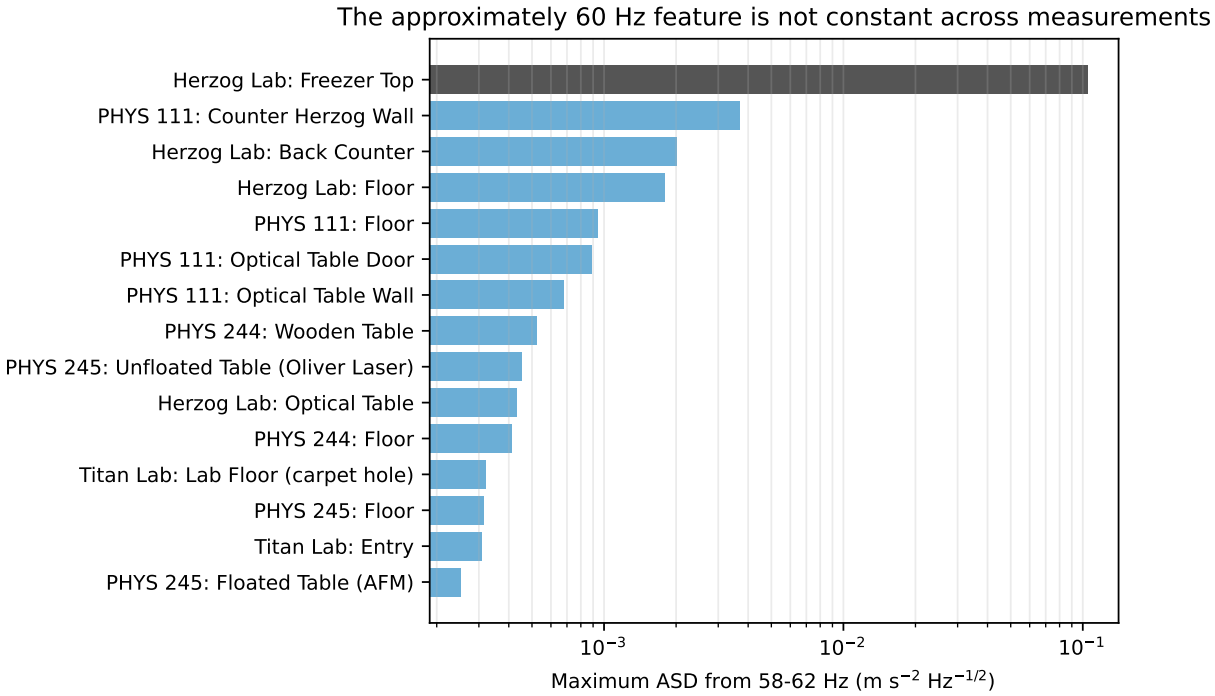


Figure 7: Maximum ASD within the excluded 58.0 Hz to 62.0 Hz band. The line appears everywhere but varies substantially in amplitude.

The universal narrow line is consistent with electrical mains coupling into the sensor or acquisition electronics. Its large run-to-run variation, especially on the freezer and surfaces near Herzog, suggests that the measured feature is not a perfectly constant electronic offset. It may contain a mechanical component or equipment-dependent electrical coupling. Either way, it is not used to rank the rooms.

9 Practical conclusion

1. **Titan Lab** is the preferred room for spectroscopic work because it has the lowest broadband background and the simplest peak structure.
2. **PHYS 245** is the best alternative. Its broadband level is close to Titan, but its 17.6, 29.8, and 66.0 Hz lines deserve attention.
3. **PHYS 111 and Herzog** have much stronger 29.8 Hz motion. Freezer-specific correction does not change that conclusion.
4. Use a floated table, but verify that it is actually floating and correctly loaded. The supplied PHYS 245 floated-table run shows that an isolation system can perform poorly near its resonance.
5. Before final installation, repeat simultaneous floor/table measurements in Titan and PHYS 245, and repeat the Herzog/PHYS 111 measurements with the freezer compressor deliberately on and off.

The measurements support a clear room choice, but they do not identify the neighboring turbine as a unique source. A synchronized on/off test would be required for that attribution.